9 **(a)** Figure 9.1 shows a copper cylinder placed on two metal rods in the magnetic field between the poles of a magnet.

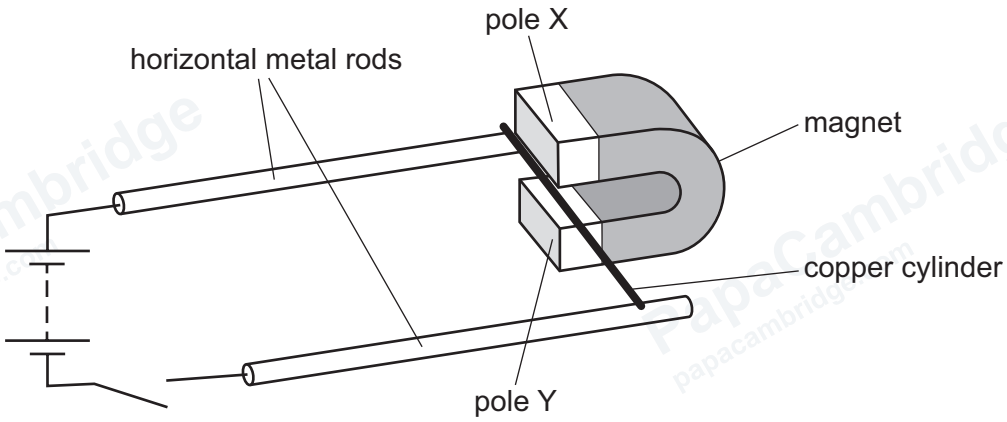

Figure 9.1

The switch is closed.

(i) On Figure 9.1, mark with an arrow the direction of the current in the copper cylinder. [1]

(ii) The copper cylinder moves to the left along the horizontal metal rods.

Complete the sentences about pole X and pole Y.

Pole X is a ..... pole.

Pole Y is a ..... pole.

[1]

(iii) Suggest **two** changes that each increase the force on the copper cylinder.

1 .....

2 .....

[2]

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(b) Figure 9.2 shows a d.c. motor.

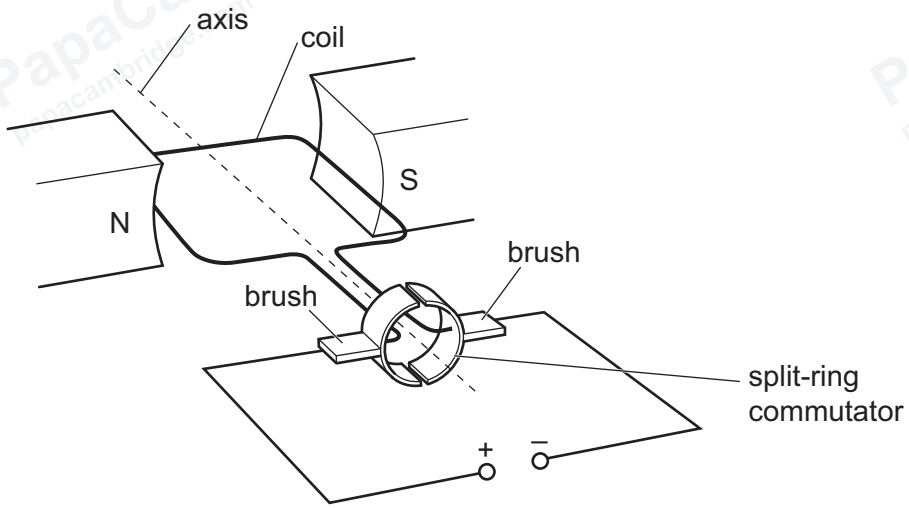

Figure 9.2

(i) Explain how the split-ring commutator allows the coil to rotate continuously.

.....

.....

..... [2]

(ii) Explain the purpose of the brushes.

.....

..... [1]

[Total: 7]

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